When and Where You Nudge Matters: Multi-Step Interface Interventions Against Address Poisoning in Crypto Wallets

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Abstract

Crypto wallets are the primary applications through which users interact with Web3 ecosystems, and are subject to a range of attacks that lead to serious security consequences. Address poisoning is a prominent example: attackers exploit wallets' truncated address displays and users' habitual reliance on simplified transfer workflows to trick them into sending funds to fraudulent addresses. While prior work has quantified the prevalence and financial impact of such attacks, little is known about how to effectively defend against them. In this work, we take wallet interface design as a practical defense vector, introducing nudge interventions that help users resist these deceptions. This approach can be deployed through wallet updates without modifying underlying blockchain protocols, and directly targets the interaction point where the attack takes effect. We design nudges targeting three key stages of the transfer workflow where users are most vulnerable to this interface-induced threat. Through a between-subjects online experiment (n = 351) and semi-structured interviews (n = 20), we show that nudges can significantly reduce susceptibility to address poisoning attacks, but that their effectiveness varies depending on where and how they are placed within the transfer flow. Our findings offer design implications for building safer wallet interfaces, demonstrating how the placement and framing of interactive elements within a transaction workflow can meaningfully shape users' security-critical decisions.

CCS Concepts

• **Human-centered computing** → **User studies**;

Keywords

Nudge Intervention Design, Address Poisoning Attacks, Web3 Security, Cryptocurrency Wallets, User Behavior

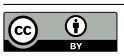
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1 Introduction

Crypto wallets sit at the intersection of users and blockchain systems, serving as the applications through which all routine Web3 interactions take place—from managing digital assets to executing transfers and engaging with on-chain protocols [18, 22, 41].[1] The interface design of these wallets is critical, as it directly shapes how users perceive and verify transaction information, and flaws in this design can be exploited by attackers [46, 51].

Address poisoning is a prominent example of such exploitation and has become one of the most serious security threats in the Web3 ecosystem [12, 20, 43], with peak activity reaching approximately 50 poisoning transfers per block on Ethereum [43]. Blockchain addresses are long, non-semantic strings that are impractical for

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[1]We provide a detailed introduction of crypto wallets in Figure 4 (Appendix A).

users to memorize or manually verify [20]. To accommodate limited screen space, wallet interfaces typically display only truncated representations—showing the first and last few characters of an address. For example, the full address shown in Figure 1(b) is displayed as `0x5f66...45d1` in the transaction history in Figure 1(a). Wallet interfaces also support copying addresses directly from transaction history (see Figure 1(b)), allowing users to locate a familiar entry and paste the address into the transfer field without manually typing the full string [20, 21, 43]. Attackers exploit this combination of truncated display and copy-from-history functionality by generating a fraudulent address that shares the same visible prefix and suffix as a legitimate recipient, and sending a transaction from this address to the victim. This transaction then appears in the victim's wallet history alongside legitimate ones. As illustrated in Figure 1(d), when the victim next initiates a transfer, the fraudulent address appears visually indistinguishable from the intended one under the truncated display. The victim may therefore mistake the poisoning transaction for the legitimate one in Figure 1(a), and then copy the fraudulent address and use it in a subsequent transfer, as shown in Figure 1(e)–(f), thereby misdirecting the funds.[2]

Since address poisoning ultimately succeeds at the interface level—where users see, select, and act upon address information through crypto wallets—no amount of backend detection or protocol-level filtering can fully protect users if the interface itself continues to present misleading information. The defense must therefore operate at the same layer as the attack: the wallet interface. To this end, we draw on the concept of *nudges*—user-centered interventions that reshape how choices are presented within an interface to guide users toward safer behavior, without restricting their autonomy [1, 16, 45, 52]. Compared to protocol-level or cryptographic defenses, interface-level interventions offer a distinct practical advantage: they can be deployed through wallet application updates without modifying underlying blockchain protocols, and they directly target the interaction layer where the attack takes effect. However, applying nudges to this setting is not straightforward. Prior nudge designs in security contexts have primarily targeted scenarios where users face a single, identifiable risky decision—such as visiting a malicious website or opening a suspicious email [52]. In address poisoning, users do not face a single risky decision, nor does any single action alone lead to harm. Instead, they proceed through a sequence of individually routine actions—selecting a transaction, copying an address, confirming a transfer—each of which appears innocuous on its own, but which collectively leads to severe and irreversible financial loss. This raises a fundamental design question: where and how should nudge interventions be placed when the threat is distributed across multiple steps, none of which appears dangerous in isolation?

To investigate how nudges can be effectively designed for this setting, we embed interventions into three key interaction points in the wallet's transfer workflow—the specific moments where users encounter and act upon address information: selecting a transaction from history, copying addresses from past transactions, and confirming a transfer request. Prior research has shown that address poisoning transactions can be identified using features such as transaction patterns and address similarity [20], enabling the implementation of interventions. Each intervention targets a distinct interface component, reshaping how address information is visually presented or how users interact with it at that stage. After an initial design phase and iterative refinement through an expert workshop, we finalized three nudge strategies: *Visual Suppression* during transaction selection, which visually de-emphasizes fraudulent transactions sent by attackers; *Copy Warning* at address copying, which triggers a warning when users attempt to copy a suspicious address; and *Confirmation Disruption* at transaction submission, which interrupts the routine confirmation process and prompts users to reverify the recipient address. We evaluated these interventions in a between-subjects controlled experiment ($n = 351$) conducted on Prolific [37], with five conditions: three single-nudge groups, one combined-nudge condition integrating all three, and a control group with no intervention. We collected post-experiment survey responses from all participants and conducted follow-up semi-structured interviews ($n = 20$) to gain deeper insights into participants' perceptions and decision-making processes.

We found that all three interventions significantly increased the proportion of participants who successfully avoided address poisoning compared to the *Control* (23.8%), with *Visual Suppression* (94%) outperforming the other two nudges. The *Combined Intervention* (91.4%) condition achieved performance comparable to *Visual Suppression*. Mechanism-level analyses revealed that *Visual Suppression* was particularly effective because it prevented users from encountering or selecting fraudulent transactions, thereby reducing the likelihood of entering the attack sequence. In contrast, *Copy Warning* and *Confirmation Disruption* successfully interrupted risky actions, but many participants still failed to distinguish legitimate addresses from fraudulent ones and subsequently repeated unsafe behaviors that resulted in transferring funds to the attacker.

Our study makes two contributions. First, we contribute to the understanding of how nudge interventions should be designed when security threats are distributed across a sequence of routine interactions rather than concentrated at a single decision point. Building on prior UI intervention guidelines [3] that security interventions should be presented at the time and place where users encounter risk [4, 32, 36], we extend this perspective to a multi-step wallet transfer workflow. We show that where a nudge is placed within a multi-step workflow matters more than whether a nudge is present, and that early-stage interventions that reshape what users see are more effective than late-stage interventions that ask users to reconsider what they have already done. These findings offer generalizable design implications for embedding security support into interactive systems where harm arises not from a single dangerous action, but from a sequence of individually benign ones. Second, we demonstrate that interface design is a powerful and underutilized lever for mitigating security threats in Web3. Our work shows that carefully designed interface interventions alone can substantially reduce user susceptibility, reframing wallet interface design not merely as a usability concern but as a first-class security mechanism.

2 Related Work

In this section, we review two bodies of work that inform our study. We first survey research on nudge-based interventions designed to

[2] A detailed explanation of how address poisoning works is provided in Appendix B.

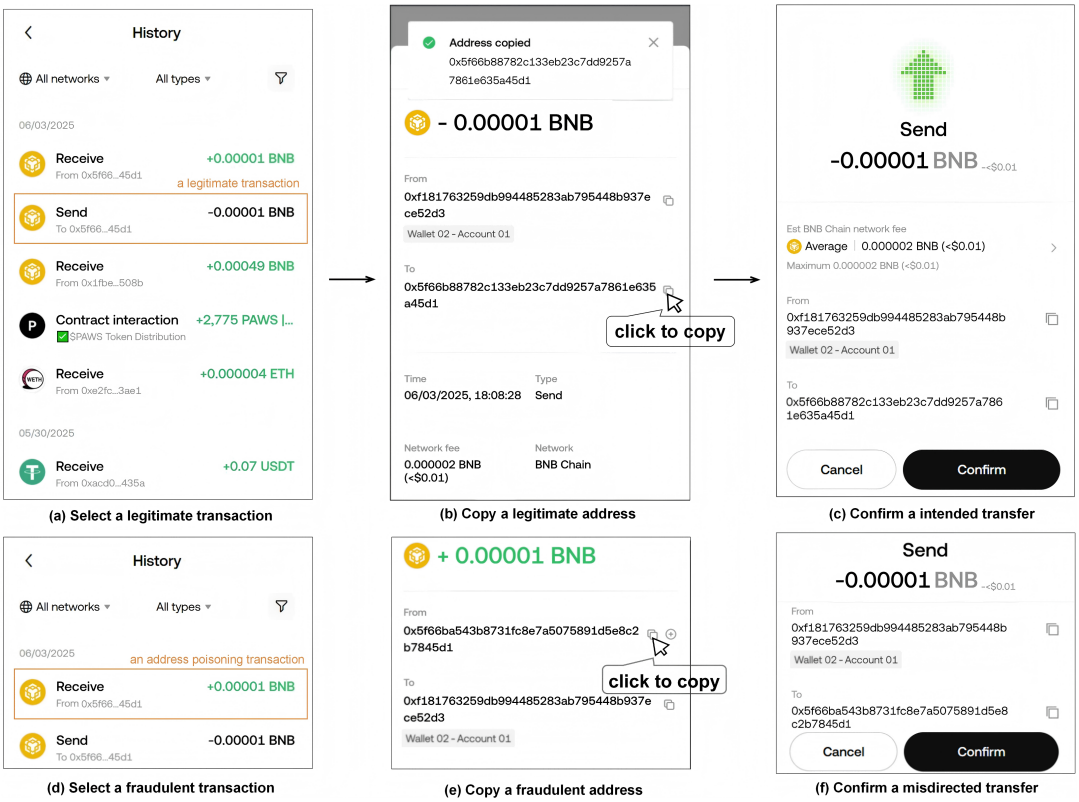


Figure 1: Victim behavior during an address poisoning attack. The top row, (a)–(c), shows the intended workflow: the user identifies a past recipient in the transaction history from a truncated address, copies the address, and confirms the transfer. The bottom row, (d)–(f), shows how a victim may mistake a poisoning transaction with an identical truncated address for the legitimate one in (a), copy the fraudulent address, and transfer funds to it.

influence users' security-related actions, with a focus on how interface design has been used to guide safer behavior. We then review existing research on address poisoning attacks, highlighting the gap between technical detection efforts and user-facing defenses.

2.1 Nudges for Security-Related Actions

A growing body of research has explored how interface design can serve as a nudge to guide users toward safer behavior in security contexts. These interventions range from visual cues and warnings to workflow-level modifications, all aimed at reshaping how users perceive risk and act upon it without restricting their autonomy.

One well-studied form of such nudging is security warnings. Bravo-Lillo et al. [10] showed that security warnings are often ineffective when their design does not align with users' mental models of risk. Akhawe et al. [2] conducted large-scale field studies demonstrating that warning effectiveness varies widely depending on warning type and user experience. Silic et al. [39] found that visual properties such as color significantly influence users' attention and compliance with warnings. Petelka et al. [36] showed that the placement of warnings within an interface also affects how users perceive and respond to them. Mengascini et al. [31] extended this line of work to immersive environments, identifying multiple nudge modalities including visual cues, object transformations, and pop-up alerts, and showing that more intuitive visual cues often outperform traditional pop-ups.

In financial domains, prior work has investigated how interface-level nudges can promote safer or more deliberate decision-making. Story et al. [40] designed interventions based on Protection Motivation Theory and implementation intentions, showing that such interventions can significantly increase real-world adoption of secure mobile payments. Marforio et al. [29] revisited personalized security indicators in mobile banking and found that their effectiveness depends on context and interface characteristics. Mutyavaviri et al. [33] examined designs for discouraging e-banking fraud, demonstrating that empathy-driven messages and system-monitoring reminders are especially effective. Chaudhry et al. [11] proposed design guidelines for promoting more deliberative investment decisions, grounded in behavioral theory.

Prior nudge designs in security contexts share a common structural assumption: they intervene at a single, identifiable moment where the user faces a clearly risky decision—such as clicking a malicious link [36], approving a suspicious transaction [40], or entering a dangerous environment [31]—or at different moments to influence the same underlying risky action [33]. Address poisoning fundamentally differs from these scenarios. Users do not face a single moment of obvious risk, nor does any single action alone

lead to harm. Instead, they proceed through a sequence of individually routine actions—selecting an entry, copying an address, and confirming a transfer—none of which appears dangerous on its own, but which collectively lead to irreversible financial loss. This creates a design challenge that prior work has not addressed: how should nudge interventions be distributed across a multi-step workflow when the threat itself is distributed, and no single step can be identified as *the* risky decision?

2.2 Address Poisoning Attacks

Address poisoning has drawn growing academic attention due to its covert techniques and severe financial impact. Prior work has primarily advanced along two directions: characterizing attack strategies and measuring their prevalence. On the attack side, studies have categorized address poisoning into multiple types—including zero-value transfers, dust-value transfers, and fake token transfers—and have shown that attackers continuously refine their methods to improve stealth and efficiency [12, 20, 43]. On the measurement side, researchers have developed detection frameworks and pipelines to identify poisoning attempts at scale, quantify financial losses, and profile attacker behavior [12, 20, 43]. However, this body of work remains focused on the attacker and system side, with little attention to how users can be protected through wallet interface design [21]. Our work addresses this gap by treating address poisoning not only as a detection problem but as an interaction design problem, and investigating how nudge interventions embedded in wallet interfaces can reduce user susceptibility.

3 Interface Designs for Attack Prevention

The interface design process began with an initial set of design goals derived from the characteristics of address poisoning attacks and insights from prior work. We then invited domain experts to discuss these goals and the corresponding design ideas in a workshop, and designed the interventions based on their feedback.

3.1 Design Goals

Following our review of prior work and the characteristics of address poisoning attacks, we set the design goals (DGs) for interface interventions in crypto wallets as follows:

DG1: Provide protection throughout sequential actions. As presented in Figure 1, completing a transfer involves three sequential actions. Therefore, our first design goal is to establish continuous protection across the interaction flow rather than at a single stage. By providing complementary interventions at different stages, the interface can offer layered protection and support users as risk accumulates, thereby reducing the likelihood of a successful attack.

DG2: Help users avoid engaging with fraudulent addresses. A key reason why address poisoning succeeds is that users select a fraudulent address as the transfer destination and continue interacting with it as if it were legitimate. Therefore, our second goal is to help users avoid engaging with fraudulent addresses. Rather than relying solely on users to detect subtle differences on their own, the interface should help them distinguish intended addresses from deceptive alternatives before the transfer is carried forward.

3.2 Expert Workshop for Design

3.2.1 Participants. We recruited five participants through targeted outreach. Participants included developers, auditors, and researchers with professional expertise in blockchain technologies. Their demographic information is reported in Table 1 in Appendix C.

3.2.2 Procedure. The workshop was semi-structured, conducted across two 2.5-hour sessions, and organized into three phases. In the introduction phase, we established common ground by briefing all participants on the study goals and procedures. In the second phase, we facilitated an open discussion of the current prototype. The researchers first explained how each goal was defined and presented the corresponding initial design space. Experts then reflected on whether the designs aligned with real-world wallet use patterns and identified potential concerns, such as message clarity, interaction burden, and the risk that users might overlook or misinterpret warnings. In the third phase, we invited experts to discuss additional opportunities for intervention across different stages of the transfer process. Throughout the workshop, we took detailed notes and iteratively compared feedback across sessions to identify recurring concerns and design suggestions.

3.2.3 Findings. In the expert workshop, nearly all experts mentioned that simply warning users about potential risk is insufficient. They emphasized that warning messages should not remain vague, but should instead provide concrete guidance on how to verify the address. As E03 explained: *"The wallet should not only tell users that an address is risky, but also tell them how to check it. For example, it should clearly prompt them to verify the full address rather than only the beginning and the end. Only then is it more likely that they will avoid choosing the wrong one again"* (E03). This finding suggests that risk communication alone is not enough; users also need explicit support in translating risk awareness into effective verification behavior. This aligns with the EPPM [48], which highlights the importance of self-efficacy and actionable guidance in motivating protective behavior. Based on this feedback, we introduced an additional design goal after the expert workshop.

DG3: Provide clear guidance for verification. The third goal is to ensure that warning messages provide actionable guidance for verifying the destination address. Rather than only alerting users to risk, the interface should direct them to check the full address instead of relying on partial character matching.

3.3 Nudge Intervention Design

Based on the wallet transfer workflow identified through our expert workshop (Section 3.2), we designed three nudge interventions, each targeting a distinct interaction point where users are vulnerable to address poisoning: selecting a transaction entry from history, copying an address from a transaction detail page, and confirming a transfer. Notably, our interventions use a binary risk representation, flagging transactions as suspicious or leaving them unflagged, as our focus was on the timing and placement of interventions within the workflow. We present each intervention in detail below.

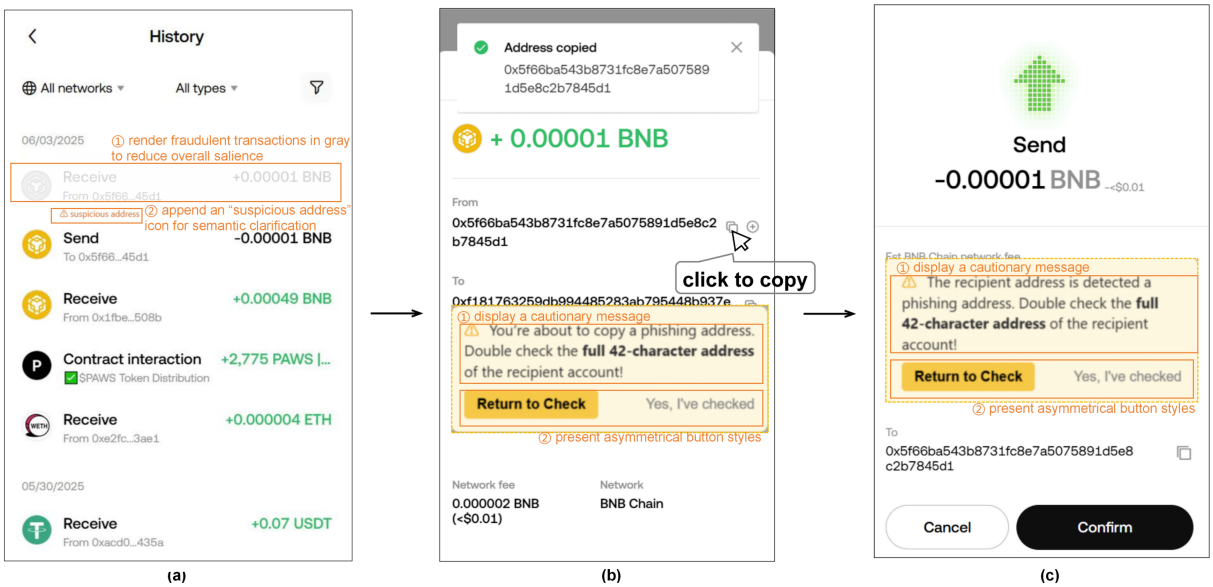


Figure 2: Overview of interventions targeting detected suspicious addresses. (a) Visual Suppression at transaction selection, which visually suppresses suspicious records using a grayed-out style with an orange "suspicious address" icon. (b) Copy Warning at address copying, a just-in-time pop-up warning that prompts users to recheck the copied suspicious addresses. (c) Confirmation Disruption at transaction confirmation, a just-in-time pop-up warning that prompts recipient verification.

3.3.1 ***Visual Suppression at Transaction Selection.*** The first nudge targets the earliest interaction point in the transfer workflow: browsing transaction history and selecting an entry. When a user's transaction history contains fraudulent entries planted by an attacker, the risk begins the moment the user selects one of these entries. We designed a *Visual Suppression* intervention that renders suspicious transactions in a grayed-out style within the transaction history view (see Figure 2a). Grounded in status quo bias [38], this design reduces the visual salience of potentially fraudulent transactions, making users less likely to select them while naturally directing attention toward legitimate records. An orange label serves as a complementary indicator to clarify why the entry is visually suppressed.

Importantly, we chose to visually de-emphasize rather than completely hide suspicious transactions. Transaction history serves as a critical financial record, and its completeness and integrity are essential for account management and auditing [26, 28]. Users are particularly sensitive to missing or altered records in contexts involving their financial assets, and removing entries—even fraudulent ones—could erode trust and undermine users' sense of control over their own accounts. *Visual suppression* preserves this integrity while still reducing the likelihood that users select and act upon fraudulent entries, preventing them from entering the attack sequence and proceeding to subsequent steps such as copying a fraudulent address or confirming a misdirected transfer.

3.3.2 ***Copy Warning at Address Copying.*** The second nudge targets the moment when a user attempts to copy an address from the detail page of a fraudulent transaction. This action reflects a deliberate intention to reuse the address for a transfer, making it a critical intervention point. We designed a *Copy Warning* intervention that is triggered when a user initiates a copy action from the detail page of a suspicious transaction (see Figure 2b). The intervention presents a warning modal that interrupts the copy action and prompts the user to reconsider. The modal flags the address as suspicious and features an asymmetric button design: a visually prominent "Return to Check" option and a less salient "Yes, I've checked" option. By introducing friction at this step, the intervention encourages users to pause and verify the address before proceeding [4, 5, 36]. This design is also related to the message-based interventions reviewed in prior self-efficacy research [6]. Its concrete rechecking guidance is intended to support users in carrying out the recommended protective action.

Importantly, the intervention does not prohibit copying. Copying an address is a self-initiated action that may serve multiple purposes and does not necessarily indicate that a user is being deceived [26]. Blocking it outright could trigger reactance and undermine trust in the wallet. Instead, *Copy Warning* preserves user autonomy while encouraging verification, reducing the likelihood that users proceed with reusing a fraudulent address.

3.3.3 ***Confirmation Disruption at Transfer Confirmation.*** The third nudge targets the final step in the transfer workflow: confirming the transaction. This is the most consequential interaction point, as submission is irreversible and directly results in financial loss if the recipient address is fraudulent. We designed a *Confirmation Disruption* intervention that is triggered when the user attempts to submit a transfer to a suspicious address. The intervention presents a warning modal that interrupts the submission flow and prompts the user to re-examine the recipient address (see Figure 2c). By introducing friction into an otherwise routine

confirmation step, the intervention encourages users to pause and reverify their action before it becomes irreversible [5]. Similarly, the rechecking guidance directs users to return and verify the full address rather than merely communicating risk [6].

Importantly, the intervention does not block the transaction. Any detection mechanism is subject to false positives—for example, two legitimate addresses may share similar prefixes and suffixes. Enforcing hard blocks in such cases could undermine user trust in the wallet [36]. Instead, *Confirmation Disruption* preserves user autonomy while providing a final opportunity for verification at the moment of highest consequence.

4 Evaluation

We conducted two evaluation studies to assess the effectiveness of the proposed interventions: (1) a controlled between-subjects experiment ($n = 351$) conducted on the online platform, and (2) follow-up interviews ($n = 20$) to gain deeper insights into participants' interpretations of the interventions. Our study was approved by our institution's ethics review board.

4.1 Study 1: Controlled Experiment

We conducted a between-subjects experiment to examine whether user behavior changes under interface interventions against address poisoning. We designed a simulated crypto wallet interface that included core functionalities such as browsing transaction history, copying wallet addresses, and initiating token transfers.

4.1.1 Experiment Design. We conducted the experiment via Prolific, an online platform with a diverse international participant pool [37]. To avoid priming participants about address poisoning, we used deception and described the study as an evaluation of crypto wallet functionality rather than an intervention experiment.

We defined five experimental conditions: a control condition with no intervention, three single-intervention conditions corresponding to the three interface interventions, and one combined-intervention condition. After accepting the study, participants were randomly assigned to one of these five conditions.

In each condition, participants completed the study in three steps. First, participants received a tutorial introducing the simulated wallet interface and the transfer procedure. Second, participants completed ten transfer tasks on the experimental platform, including three benign scenarios and seven address poisoning scenarios. The seven poisoning scenarios were constructed by adapting and combining representative attack characteristics identified in prior empirical studies of address poisoning [43], including small-value or zero-value transfers, token mismatches, and repeated poisoning attempts. The scenarios varied in characteristics such as token type and transfer amount, representing realistic poisoning contexts in transaction histories. This design was intended to examine how users respond to different attack characteristics, and the ratio of benign transfers to attack scenarios was consistent with prior attack studies that included both benign and adversarial tasks [36]. Task order was randomized to mitigate order effects. Third, after completing the transfer tasks, participants completed a survey assessing perceived task difficulty, confidence in transfer correctness, and their perceptions of the interface interventions (see Appendix D.5 for the full survey). At the end of the study, we debriefed participants and informed them that the true purpose of the study was to evaluate the interface interventions.

4.1.2 Pilot Study. Before the formal experiment, we conducted a pilot study with 10 participants to validate the study design and gather feedback for refinement. Based on the pilot feedback, we refined the presentation of task-related information in the interface to improve clarity. Further details are provided in Appendix D.2.

4.1.3 User Recruitment and Demographics. We recruited participants via Prolific and randomly assigned them to one of five conditions: a control condition with no intervention, three single-intervention conditions corresponding to the three interventions, and one combined-intervention condition. Eligibility was restricted to participants with confirmed English fluency to minimize comprehension issues. In total, 413 participants were recruited.[3] Participants received £2.50 for a study expected to take 20–25 minutes, which was above Prolific's recommended minimum hourly wage.

After data cleaning, which excluded 62 incomplete or invalid responses, the final sample consisted of 351 participants. Of these, 151 identified as male (43.1%) and 200 as female (56.9%). Participants ranged in age from 18 to 76 years ($M = 33.65$, $Md = 30$). Each condition included 64 to 78 participants. Participants also varied in prior crypto wallet experience: 179 reported no prior use, 73 reported less than one year of experience, 72 reported 1–3 years of experience, and 27 reported more than three years of experience.

4.1.4 Data Collection and Analysis. The data used for analysis included detailed behavioral logs, such as the time spent on each interface screen, the sequence and frequency of button clicks, the recipient address entered at transaction confirmation, and responses to the post-test survey. We excluded 62 responses with missing interaction logs due to system errors or transfers sent to unrelated, non-task addresses. The final sample consisted of 351 participants.

To evaluate the effectiveness, we combined task outcome analysis with behavioral process analysis. We constructed models using Generalized Estimating Equations (GEE) with a binomial logit link [27], reporting differences across conditions using Wald χ^2 tests based on this model [47]. First, to assess overall effectiveness, we examined task correctness, which indicates whether participants avoided transferring funds to a fraudulent address and serves as the primary outcome measure. Along with Wald χ^2 test results, we also reported odds ratios (OR) and risk differences (RD) as effect sizes [17] to quantify the magnitude of each intervention effect.

Secondly, to analyze differences in stage-specific binary behaviors such as first-click behavior, we used the behavioral outcome as the dependent variable to assess whether the specific behavior differed across conditions. Specifically, to evaluate whether these behaviors predicted final task correctness, we used Firth's bias-reduced logistic regression [14]. Finally, we reported Wald χ^2 test results for heterogeneity, testing whether intervention effects varied across participant characteristics (e.g., Web3 experience), task features, and repeated exposure over time.

For the post-test survey, we analyzed three components: scale-based items, open-ended responses, and background information.

[3]The final sample size exceeded the minimum number of participants required by the *a priori* power analysis conducted in G*Power. Details of the sample size determination are provided in Appendix D.3.

For scale-based items measuring perceptions such as transaction confidence, we computed descriptive statistics and used Kruskal–Wallis tests [25] to compare conditions, treating the single 5-point Likert item as an ordinal outcome. We analyzed open-ended responses using thematic analysis [7]. Two researchers coded the responses independently and reached approximately 87% initial agreement before resolving minor discrepancies through discussion. Given this high level of agreement, we did not conduct a formal inter-rater reliability test [30, 35]. We then performed spot checks to ensure consistency and consolidated related codes into broader themes. Finally, we combined background survey data with experimental performance and used GEE logistic regression [27] to examine whether prior experience influenced task correctness and behavioral responses.

4.1.5 Limitations. Our study has two main limitations: the simulated setting and participant demographics. First, our study faces an ecological validity challenge common to experimental research on security behaviors [36]. To mitigate this limitation, we adopted several established strategies. Specifically, we used deception by not disclosing the attack focus of the study to reduce expectancy effects [42], and approximated real-world interaction contexts [24] by implementing a high-fidelity prototype wallet interface aligned with mainstream crypto wallets. These approaches are consistent with prior work and security usability studies and have been shown to improve behavioral realism [24, 42]. Second, our sample was skewed toward novice users, with nearly half reporting no prior wallet experience. However, address poisoning is typically conducted at scale [20], with attackers exploiting broadly shared behavioral heuristics rather than selectively targeting users by experience level. Moreover, our heterogeneity analysis found no significant differences in task accuracy or responses to interventions between novice and experienced participants. Together, these results suggest that the higher proportion of novice users is unlikely to affect the validity or generalizability of our findings.

4.2 Study 2: Follow-up Interviews

The goal of the follow-up interviews was to gain a deeper understanding of the underlying behavioral mechanisms by examining participants' perceptions of the interface interventions and how these interventions shaped their decision-making.

4.2.1 Participant Recruitment. After the experiment, participants were asked in the survey whether they were willing to take part in a follow-up interview. From those who consented, we recruited 20 participants for semi-structured interviews. Participants were selected based on their task performance, demographic characteristics, and prior experience with crypto wallets. The mean age of the interview participants was $M = 31$; 12 identified as male and 8 as female. We recruited approximately four participants from each of the five conditions in Study 1. Their demographics are reported in Table 3 in Appendix E.1.

4.2.2 Interview Procedure. Prior to each session, we obtained informed consent for recording and anonymous quotation. Each interview lasted 35–45 minutes, and participants received a $10 honorarium. The interviews were semi-structured and organized into four parts: (1) participant background and prior Web3 experience; (2) decision-making processes during the tasks and responses to nudges; (3) comparisons between Web3 and traditional financial transfers; and (4) evaluations and suggestions regarding the interventions. The interview protocol is provided in Appendix E.2.

4.2.3 Data Collection and Analysis. We transcribed all interviews and analyzed them using inductive thematic analysis [8], allowing codes and themes to emerge from the data rather than relying on predefined categories. Two researchers independently open-coded 20% of the transcripts, discussed and merged overlapping codes, and iteratively refined a shared codebook, reaching high agreement (Cohen's $\kappa = 0.84$); one researcher then applied the finalized codebook to the full dataset. The complete codebook is provided in Appendix E.3. We reached data saturation when no new codes emerged after P18 [9]. Given the qualitative nature of the study and the small sample size, we do not report exact counts in Section 5, and instead use the following terminology to provide a relative sense of frequency of major themes [13]: a few (0-25%), some (25-50%), many (50-75%), and almost all (75-100%).

5 Results

We structure our results in two parts: (1) the overall effectiveness of the interventions on participants' task correctness across conditions, and (2) an analysis of how the interventions shaped user behavior at each interaction stage. We also conduct a heterogeneity analysis and find that, aside from a decline in responsiveness over repeated exposures, neither participants' Web3 experience nor differences across address poisoning scenarios led to variation in the intervention effects (details in Appendix F.3). An overview of our findings is presented in Figure 8, and detailed statistical tables are reported in Appendix F.4. To simplify presentation in this section, we refer to *Visual Suppression*, *Copy Warning*, *Confirmation Disruption*, and *Combined Intervention* as *VS*, *CW*, *CD*, and *CI*, respectively.

5.1 Overall Effectiveness of Interventions

All interventions improved transfer correctness. To evaluate the overall effectiveness of the three interface interventions, we analyzed task correctness rates, defined as the proportion of attack tasks in which participants successfully transferred funds to the legitimate address. The intervention with the highest correctness rate was *VS* (94.0%), followed by *CI* (91.4%), *CD* (62.5%), *CW* (41.0%), and the *Control* condition (23.8%). The Wald χ^2 tests further showed that all interface interventions significantly improved correctness relative to the *Control* condition (all $p < 0.05$; see Table 4). Effect size estimates indicated that *VS* and *CI* had the strongest effects (70.0 and 67.2 percentage-point improvements over *Control*, respectively), followed by *CD* (38.4 points) and then *CW* (17.0 points). We also observed significant differences among the intervention conditions. *VS* and *CI* outperformed the other two intervention conditions, and *CD* significantly outperformed *CW*.

Participants reported high confidence in transfer correctness. In the post-test survey, we asked participants how confident they were that they had sent all transactions to the correct addresses. The mean confidence score was at least 4.5 on a 5-point scale across all conditions (see Table 5). A Kruskal–Wallis test showed no significant differences across conditions ($p > 0.05$), indicating that

participants reported similarly high confidence regardless of the intervention they received. Notably, this high level of confidence persisted even when participants had mistakenly sent funds to attacker-controlled addresses. This suggests that many users were unaware of their mistakes and remained convinced that they had acted correctly. The finding further highlights the importance of interface interventions, as errors that go unnoticed are unlikely to be identified and corrected by users themselves.

5.2 Behavioral Mechanisms of the Interventions

Beyond evaluating the overall effectiveness of the interface interventions, we further examine the behavioral mechanisms through which they helped users avoid address poisoning attacks. Specifically, we analyze user interactions at the three stages: selecting transactions, copying addresses, and confirming transactions. At each stage, we assess whether the intervention altered user behavior and whether these behavioral changes were associated with successful attack avoidance. Figure 3 summarizes these effects.

5.2.1 ***Stage 1: Transaction Selection.*** In address poisoning, the attack sequence begins when users select a fraudulent transaction from their history. *VS* intervenes at this earliest interaction point, aiming to direct users away from fraudulent entries and reduce the likelihood of initiating the subsequent steps of the attack sequence.

Visual suppression promoted safe initial selection and improved task correctness. To evaluate whether the intervention influenced transaction selection, we tested whether participants in the *VS* condition were more likely to first select a legitimate transaction compared to other conditions. We defined a binary variable `FirstClickSafe`, indicating whether the participant's first selection was on a legitimate entry, and compared rates across conditions using Wald χ^2 tests. The *VS* group achieved the highest rate (93%), compared to approximately 20% in other individual intervention conditions, with all pairwise differences statistically significant (all $p < 0.05$; see Table 6). This shows that *VS* substantially increased the likelihood of selecting a legitimate transaction on the first attempt. Notably, since *CI* incorporates *VS* as its first intervention component, its rate (90%) showed no significant difference from *VS*.

We further examined whether initial selection predicted final task correctness. Using Firth logistic regression to address separation issues, we modeled `Correctness` as a function of `FirstClickSafe`, showing a strong association between safe initial selection and final task correctness ($p < 0.05$): once participants made a safe first selection, transfer errors were almost entirely eliminated. When both `condition` and `FirstClickSafe` were included in the model, the latter remained the dominant predictor, while `condition` became negligible and nonsignificant.

Users perceived visual suppression as an intuitive risk cue. We analyzed participants' perceptions of the *VS* design to understand why it shaped behavior at the transaction selection stage. User-level verification analysis showed that 63.8% of participants avoided suppressed entries without performing any address verification. Among those who performed at least one verification, only 20% reported that the grayed-out entries signaled an address discrepancy. These findings suggest that *VS* operated primarily through interface-guided attentional steering rather than explicit security reasoning.

Consistent with this interpretation, almost all interview participants treated suppressed entries as less preferable. For instance, P08 explained that even without fully understanding the label, they were intuitively drawn to transactions that appeared more "normal". *"When choosing a transaction, I would not select this one; my intuition tells me it looks suspicious. And there were other transactions on this page that I considered safe, so I just clicked on those instead"* (P08).

5.2.2 ***Stage 2: Address Copying.*** After selecting the transaction and opening the details of a suspicious transaction, users may proceed to copy the malicious address. To interrupt this behavior, we introduced *CW*, which displays a warning modal when a user attempts to copy a suspicious address. At this point, users must decide whether to proceed with the copied address or recheck the address, and copy the correct address instead.

Copy warning prompted participants to revise copied addresses. We defined `CopyChange`, a binary variable to indicate whether the participant ultimately changed the copied address (e.g., by later copying the correct address). As shown in Table 7, 14% of participants in the *CW* condition changed their copied address, compared to 7% in the *Control* condition, and this difference was statistically significant ($p < 0.05$). No significant difference ($p > 0.05$) was observed between the *Combined Intervention* condition (10.6%) and the *CW* condition in terms of *CopyChange*, suggesting that the corrective effect was similarly preserved in the combined design.

Revising the copied address strongly predicted attack avoidance, but many participants failed to follow through. We next tested whether `CopyChange` predicted successful attack avoidance. Using Firth logistic regression, we found a strong effect: participants who changed their copied address were significantly more likely to complete the task correctly ($p < 0.05$). When both `condition` and `CopyChange` were included in the model, the effect of `CopyChange` remained dominant, while the direct effect of `condition` became negligible and nonsignificant.

To better understand participants' responses to the *CW* intervention, we decomposed their behavior into two sequential steps: (1) whether they canceled the copy action and rechecked the transaction, and (2) whether they subsequently selected and copied the correct address. In the first step, 30.4% of participants returned to recheck after encountering the warning modal, whereas 69.6% proceeded to copy the suspicious address without verification. Among those who returned, 46% ultimately selected the legitimate address. These findings suggest that although *CW* prompted some users to pause and reconsider, its ability to redirect them to the correct address was limited. Fewer than half of those who interrupted their copying behavior ultimately avoided the attack address, highlighting both the promise and the limitations of this intervention.

Limited effectiveness stemmed from low risk perception, superficial rechecking, and interaction costs. To understand why the *CW* intervention had only limited effectiveness, we analyzed how participants interpreted the interventions and how these interpretations shaped their subsequent actions.

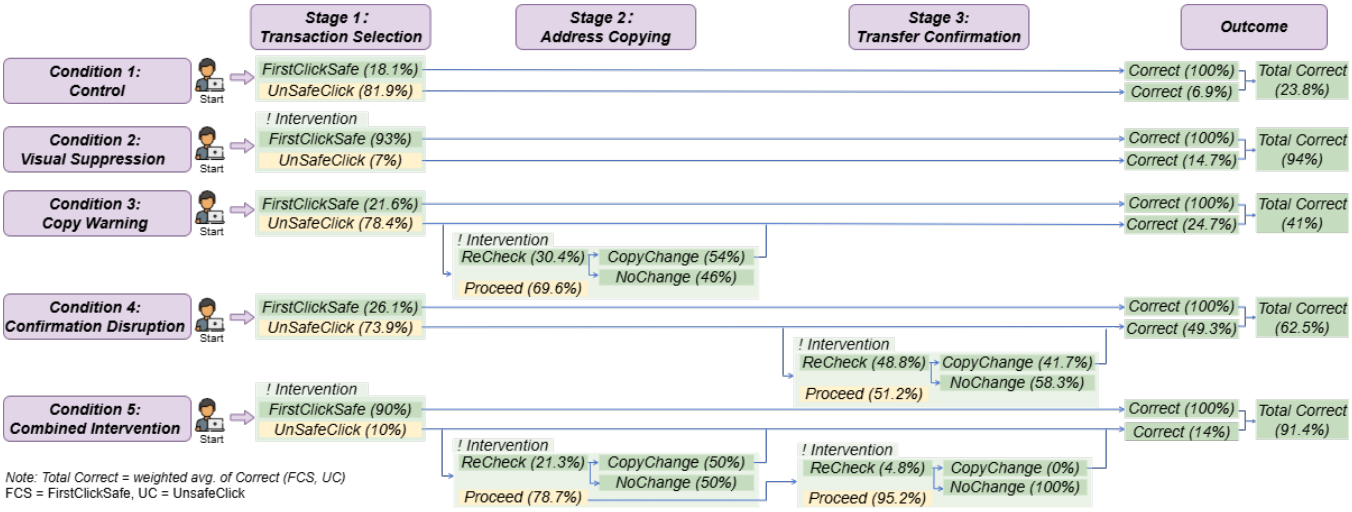


Figure 3: Behavioral flow across stages under different intervention conditions. Each row shows how users transition from selecting (Stage 1), copying (Stage 2), and confirmation (Stage 3) to the final outcome. Percentages denote proportions conditioned on the preceding behavioral step, unless otherwise noted. For UnSafeClick, Correct includes both cases without a warning (self-correction) and cases where users corrected behavior under the warning. Total Correct denotes the overall success rate in each condition, computed as the weighted sum of Correct outcomes from FirstClickSafe and UnSafeClick.

First, during the initial decision of whether to cancel or proceed, some participants noticed the warning but did not perceive copying as a high-risk action. They failed to associate the act of copying with the attack itself and instead treated the warning as a routine or overly cautious prompt. Because copying felt like a benign step with no immediate consequence, they often ignored the modal. As P04 explained: *"Since I had copied the address directly, I felt confident that no mistake could have occurred, so I simply ignored the pop-up without paying much attention"* (P04).

Second, among those who returned to recheck, many still failed to identify the malicious address as problematic, largely due to superficial verification strategies such as checking only the first and last few characters rather than the full address. As P01 explained: *"I did not fully compare it; I only briefly checked the first and last letters. I thought if the beginning and the end matched, then the middle could not be wrong"* (P01). This suggests that even when prompted, rechecking was often insufficient for detecting the deception.

Finally, many participants cited usability barriers as a reason for not selecting a new address. They noted that reselecting required too many steps—returning, rechecking, and copying again—which disrupted their workflow. As P16 put it: *"The process was so complex—you had to return, click, re-check, and finally click to copy. I felt the one I copied was correct, so I thought it was unnecessary"* (P16).

5.2.3 ***Stage 3: Transaction Confirmation.*** In an address poisoning, the final step in the behavioral chain occurs when the user confirms a transfer to a malicious address, thereby authorizing an irreversible transaction on the blockchain. To interrupt this last-stage action, we introduced the *CD* interface intervention, which displays a warning modal when a user attempts to confirm a flagged transaction. At this point, users must choose either to proceed or to cancel and recheck the transaction details.

Confirmation disruption prompted more users to return and recheck. We defined a binary variable `ConfirmModalChoice`, indicating whether the user chose to return and recheck rather than proceed. As shown in Table 8, 48.8% of participants in the *CD* condition canceled the transaction after encountering the warning. Among those who canceled, 41.7% rechecked the address and selected the correct one for the final transfer (see Table 10).

We then tested whether `ConfirmModalChoice` predicted successful attack avoidance. Using Firth logistic regression, we found a strong effect: participants who returned and rechecked were significantly more likely to complete the task correctly than those who proceeded without verification ($p < 0.05$). When both `condition` and `ConfirmModalChoice` were included in the model, the effect of `ConfirmModalChoice` remained dominant (OR $\approx 6.09 \times 10^9$), while the direct effect of `condition` was small and nonsignificant.

Confirmation disruption was more effective at interrupting risky actions than copy warning, but did not improve the quality of rechecking. Because *CW*, *CD*, and *CI* all used modal warnings at different stages of the transfer process, we compared participants' responses across these conditions. As shown in Table 8, 30.4% of participants in the *CW* condition chose to return and recheck, compared with 48.8% in the *CD* condition, and this difference was statistically significant ($p < 0.05$). This indicates that when a modal warning was presented at one stage, *CD* was more effective than *CW* in prompting users to reconsider their decision.

However, in the *CI* condition, which also incorporated *CD* as the final intervention step, only 4.8% of participants chose to return and recheck. Specifically, this drop in the CI condition is due to the small number of participants (N = 13) who reached the CD modal compared to the CD condition (N = 56). To strengthen the interpretation of CI behaviors, we further report modal reach rates

and responses across conditions (Table 9). Warning redundancy and habituation are plausible explanations, supported by our further qualitative analysis (see Appendix F.3.2), although the small sample limits stronger conclusions.

Furthermore, among those who returned to recheck, the proportion who ultimately submitted to the correct address was similar in the *CW* and *CD* conditions (46% vs. 41.7%, $p > 0.05$). These findings suggest that although *CD* outperformed *CW* in interrupting initial risky actions, it did not improve the correctness of subsequent behavior. In other words, once users chose to recheck, their likelihood of correcting the error was largely independent of the specific interface intervention that triggered the interruption. Notably, in the *Combined Intervention* condition, the proportion ultimately submitting to the correct address was 0%, suggesting that repeated prompts across stages may have undermined participants' responsiveness.

Its effectiveness was shaped by users' sense of finality, routine warning perceptions, and limited rechecking quality. We examined participants' perceptions of the warning and how these shaped their subsequent actions to better understand how the *CD* interface intervention influenced behavior.

First, compared with the *CW* condition, in which many participants regarded copying as a low-risk action, participants in the *CD* condition were generally more cautious. Although some did not fully understand why the warning was triggered, they expressed heightened concern because the confirmation step was perceived as the final and irreversible stage of the transfer process. As P12 explained: *"I felt confused and a bit panicked. And if I choose to proceed, I couldn't take it back. So I decided to return and compare the addresses"* (P12). Similarly, P18 illustrated this reasoning in a self-drawn sketch (Figure 9), showing how the warning prompted them to recheck rather than proceed because of the risk of immediate and irreversible token loss.

Second, some participants perceived the warning as routine based on prior experience with traditional financial systems. They perceived that the modal lacked urgency and failed to disrupt their intention to proceed. As P07 noted: *"I thought it was just a routine reminder, like a standard bank transfer prompt … felt it was only a standard confirmation step"* (P07).

Finally, the same user-level analysis as for *VS* showed more evident security reasoning in *CD*: 56.2% verified addresses, and 41.7% of them identified a discrepancy, similar to *CW*. When *CD* failed, however, those who returned to recheck and initiate a new transaction faced similar challenges observed in the *CW* condition. Some relied on superficial verification strategies, such as checking only the first and last characters, which limited their ability to detect the attack. Others found re-selection tedious and preferred to complete the task quickly. As P03 described: *"I felt it would be too troublesome, because if I took one more step, the transaction would be completed. At that point, the system reminded me and I checked, but found no issues, so I wasn't willing to go back and click another address again"* (P03).

6 Discussion

Based on our findings, we discuss design lessons for interventions targeting a multi-step workflow, using address poisoning as an illustrative example. Building on these lessons, we further propose design implications for strengthening interface interventions in crypto wallets.

6.1 Intervention Effects in Multi-Step Behavior

Our findings highlight two key insights: (1) interventions must go beyond interrupting incorrect behavior to guide users back onto safe paths within a multi-step workflow, and (2) the fundamental protection lies in users' security awareness of their own behavior.

Redirection drives intervention effectiveness. First, address poisoning, as a multi-step interaction example, points to a challenge in security intervention design: an intervention is effective not merely when it creates friction at risky moments, but when it also redirects users back to a safe behavioral path. Many prior interface designs treat user behaviors as a single, identifiable decision point and therefore guide safer choices mainly by examining how information is presented. Examples include PMT-based nudges [40] and color-coded warnings [39], which are typically evaluated based on users' immediate choices. However, in address poisoning, risk accumulates across multiple steps and is only realized at execution, suggesting that a single intervention point may not be sufficient to guide users away from unsafe outcomes. Instead, effective defense may require interventions that align with users' behaviors across different stages of the interaction flow.

In our study, the observed differences in effectiveness may reflect two factors. First, intervention formats matter. As shown in Section 5.2.1, VS guided users away from fraudulent transactions with limited active verification, whereas CW and CD required users to engage with warnings and accurately recheck addresses. However, many users relied on superficial verification strategies and still failed to identify malicious addresses when rechecking (Section 5.2.2). Second, an intervention's position in the interaction sequence may also affect its effectiveness. VS intervened before users committed to a suspicious selection, whereas CW and CD interrupted users and prompted them to revisit and reverse prior choices. Some participants found this process burdensome or ignored the warnings because they remained confident in their selections (Section 5.2.3). This aligns with prior work showing that interventions introduced before users commit to a selection can be more effective than those introduced afterward [15]. However, because intervention format and position were partly coupled in our design, their respective effects cannot be fully disentangled.

Regardless of their relative contributions, both explanations point to the importance of redirecting users toward a safe behavioral path. Corrective behavior in CW and CD was also closely associated with successful attack avoidance when users recovered from an initially unsafe action (Sections 5.2.2 and 5.2.3). The limitation of these interventions may therefore lie not simply in their timing, but in interrupting unsafe behavior without providing sufficient support for redirection. Thus, across intervention formats and workflow positions, effectiveness depends not only on creating friction at risky moments, but also on guiding users back to a legitimate course of action.

Risk perception shapes intervention effectiveness. Second, users' awareness is the foundation of attack defense. The effectiveness of interventions depended not only on whether users paused at risky

steps, but more fundamentally on whether they recognized those steps as risky in the first place. Many participants expressed high confidence in their actions even when incorrect, treating actions such as copying or confirming an address as routine, low-stakes operations rather than critical points of vulnerability (Section 5.1). This aligns with prior evidence [34] that overconfidence is associated with a higher likelihood of falling victim to attacks. This also highlights how cognitive biases can lead users to misinterpret interface cues [10, 31], causing warnings to be misunderstood or ignored rather than interpreted as signals of danger, and thereby undermining the effectiveness of interaction design.

In address poisoning, however, the workflow unfolds across a sequence of routine actions including selecting, copying, and confirming, where each individual step appears normal and low-risk. The absence of salient anomalies at any single step makes it difficult for users to recognize when verification is needed. As a result, even when nudges successfully interrupt an action, users often resume the workflow without correction, perceiving their prior choices as safe. This highlights how risk in multi-step workflows can accumulate subtly across otherwise benign interactions.

6.2 Design Implications

We propose two design implications for mitigating address poisoning in multi-step interactions: (1) how interface intervention designs can support behavioral recovery and (2) how interface designs can sustain risk awareness across interaction stages.

Support recovery in multi-step interactions. In multi-step interactions, nudges should not only interrupt risky actions but also actively guide users back to safe transaction paths. Interfaces can support recovery by directly linking users to the correct transaction record, rather than merely prompting rechecks. At the selection stage, wallets can suppress fraudulent entries by reducing their visual prominence (e.g., greying out), thereby guiding users toward safe initial choices. At later stages (e.g., copy or confirmation), wallets can facilitate correction by enabling one-click recovery. For example, the interface can surface the alternative candidate (the most likely legitimate target), highlight discrepancies between the selected address and the alternative one, and allow users to easily revert to it. Embedding such recovery mechanisms within the multi-step interactions ensures that interventions steer users toward legitimate actions across multiple steps, rather than relying solely on reflection or isolated warnings.

Sustaining awareness across multi-step interactions. Effective interventions for address poisoning should maintain users' risk awareness throughout the multi-step interactions by embedding step-specific cues at key interaction points. At the selection stage, wallets can highlight fraudulent entries by marking visually similar addresses (e.g., sharing the same prefix and suffix) and labeling unfamiliar ones (e.g., "no prior interaction"). At the copying stage, interfaces can display explicit usage signals for the selected address, such as interaction frequency and recency, and flag anomalies (e.g., "no prior non-zero incoming transactions from you" or "rarely used"), helping users quickly assess legitimacy. At the confirmation stage, wallets can present a concise summary comparing the selected address with frequently used alternatives, enabling users to detect inconsistencies before finalizing the transaction. By reinforcing awareness across multiple stages, interventions can help users internalize vigilance and translate brief prompts into consistent, safe behavior across the chain of interactions.

6.3 Detector Quality and Attacker Adaptation

The effectiveness of our interventions depends on the quality of the underlying address-poisoning detector. False negatives (FN) may leave poisoned addresses visible, whereas false positives (FP) may trigger unnecessary interventions. In addition, detection thresholds may introduce trade-offs between coverage and precision that affect both prompt frequency and protection effectiveness.

From a usability perspective, FP may affect usability and user trust by increasing verification effort, interrupting transaction workflows, and contributing to warning habituation over time. To isolate users' responses to intervention design, our study assumes a high-precision detector, consistent with prior work reporting no false positives in its evaluated setting [43]. However, real-world deployments may involve different precision–coverage trade-offs that shape how users respond to interventions.

Furthermore, attackers may adapt to detection strategies by exploiting fixed similarity rules or temporal windows. Existing detectors often rely on character-level similarity within a temporal window [20, 43], while our interviews showed that users inspect only a small portion of an address. Attackers could therefore craft look-alike addresses that evade fixed matching rules or delay poisoning beyond the detector's temporal window, leaving more malicious addresses unsuppressed or unwarned.

In real-world deployments, address-poisoning defenses may face challenges discussed above. In this context, examining how detector quality, usability considerations, and attacker adaptation jointly influence the effectiveness of user-centered interventions in practice constitutes an important direction for future research.

7 Conclusion

In this paper, we designed three nudge interventions tailored to users' action paths in address poisoning, a unique form of attack that exploits users' habitual address-selection behaviors during token transfers. Through a between-subjects experiment and follow-up interviews, we showed that all three interventions improved transfer correctness, and the most effective approach was visual suppression during transaction selection, which reduced users' tendency to select look-alike addresses. Our work advances the design of nudges for security threats that unfold across routine interactions and demonstrates how crypto wallet interfaces can function as security mechanisms to support safer user behavior.

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A Wallet Interface Introduction

The wallet interface consists of multiple interconnected views that support users throughout the transaction workflow, as shown in Figure 4 (a). The main wallet interface displays the user's account information, including token balances and primary actions such as *Send* and *History*. These two functions form the core interaction pathways: users can either review past transactions through the history view or initiate a new transfer through the send workflow.

Transaction History. The transaction history (see Figure 4 (b)) presents records of previous transactions associated with the account, categorized as *Send* or *Receive*. Each entry includes the counterparty address (e.g., the "To" field indicating the recipient in a sent transaction), as well as the transferred amount and token type. Users can browse these entries to identify prior interactions, and selecting a specific entry navigates to a detailed transaction view.

Transaction Details and Address Reuse. The transaction detail view (see Figure 4 (c)) provides comprehensive information about a selected transaction, including the sender address, recipient address, transaction timestamp, and execution status. This view also allows users to copy addresses directly, which serves as a common mechanism for selecting recipients in subsequent transactions. By copying an address from a past transaction, users can efficiently reuse previously interacted addresses without manual entry.

Send and Transaction Initiation. To initiate a transfer, users select the *Send* action from the main wallet interface, which opens the transaction initiation interface (see Figure 4 (d)). In this view, users specify the recipient address, select the token type, and input the transfer amount. Rather than manually entering a full address, users can paste an address previously copied from the transaction detail view, enabling efficient reuse of prior interaction records.

Transaction Confirmation. After entering the required information, users proceed to the transaction confirmation page (see Figure 4 (e)), where a summary of the transaction is displayed, including the sender, recipient, transfer amount, and associated network fee. Users must explicitly confirm the transaction at this stage before it is executed on the blockchain.

B Mechanism of Address Poisoning Attack

Figure 5 illustrates the basic mechanism of an address poisoning attack. An address poisoning attack unfolds in four stages.

Stage 1: Identifying the Victim. The attacker first monitors publicly available transactions on the blockchain to identify a potential victim (V). Because blockchain transactions are transparent and permanently recorded [49], attackers can scan data at scale to locate accounts that are active, frequently transfer tokens, or hold non-trivial balances [23, 50].

Stage 2: Generating the Fraudulent Address. Once a target is selected, the attacker generates a fraudulent address (P) that closely resembles a legitimate address (L) with which the victim has previously interacted, typically sharing the same prefix and suffix. This is achieved through vanity address generation, which iteratively produces key pairs until an address with the desired prefix or suffix is found. For example, given victim address `0x1fbe...508b` and counterparty `0x5f66...45d1`, the attacker can generate `0x5f66ba543b8731fc8e7a5075891d5e8c2b7845d1`, which shortens to the identical `0x5f66...45d1`.

Stage 3: Interacting with the Victim to Poison the History. The attacker then uses the fraudulent address to send a transaction to V, ensuring that this deceptive address P appears in the victim's wallet transaction history. This is achieved by initiating a transaction involving both the victim's account and the attacker-controlled address, thereby ensuring that the fraudulent address appears in the wallet interface. Because Web3 wallets often display only shortened address representations, showing only the prefix and suffix, the fraudulent address can be visually indistinguishable from the legitimate one during the victim's next intended transfer to L.

Stage 4: Victim Sends Funds to the Fraudulent Address. Because blockchain addresses are long and difficult to memorize, users commonly copy them from transaction history when sending funds to familiar counterparties, rather than typing them manually [20, 21, 43]. In crypto wallets, addresses are usually truncated, leaving users to distinguish them mainly by their visible prefix or suffix [19, 20]. Since the fraudulent address is crafted to mimic these visible segments, it can appear indistinguishable from the legitimate one in the wallet interface. As a result, during subsequent transactions, the victim may mistakenly transfer funds to the phishing address P instead of the intended legitimate address L [20].

C Demographics of Experts

The demographic information of expert interview participants is shown in Table 1.

Table 1: Demographic information of the experts. *Note: YoE refers to experts' years in Web3.*

ID	Age	Education	Occupation	YOE
E01	25–34	Master's	Developer	3–5
E02	35–44	Bachelor's	Developer	>5
E03	25–34	Bachelor's	Smart contract auditor	>5
E04	35–44	Master's	Wallet developer	3–5
E05	25–34	Ph.D.	Postdoctoral researcher	>5

D Supplementary Experimental Materials

D.1 Experimental Conditions

An overview of the experimental conditions, targeted interaction stages, and corresponding nudge interventions is shown in Table 2.

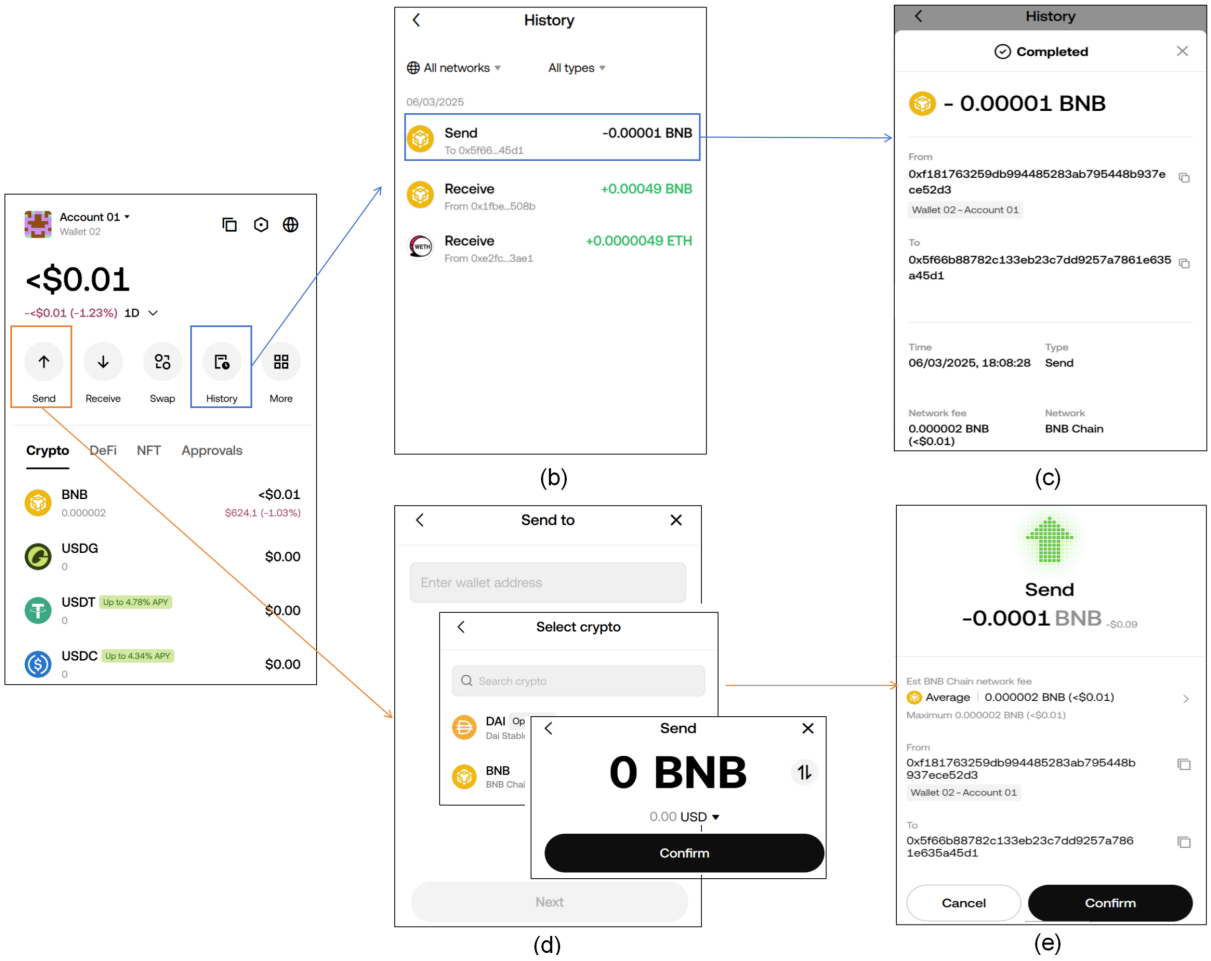


Figure 4: Example wallet interface. (a) Main interface showing account, balances, and actions (Send, History). The History button opens the transaction list (b). (b) Transaction history displaying past Send/Receive records with counterparty address, amount, and token type. Selecting an entry opens its detail view (c). (c) Transaction details including sender, recipient, time, and status; addresses can be copied. (d) Transaction initiation interface, accessed via Send, where users enter recipient, token, and amount. Addresses copied from (c) can be pasted directly, enabling quick recipient selection. (e) Confirmation page summarizing transaction details (sender, recipient, amount, fee) before final approval.

D.2 Details of Pilot Study

Before conducting the formal experiment, we carried out a pilot study to validate the feasibility of the study design. Ten participants were recruited through the authors' personal networks and randomly assigned to one of the five experimental conditions. The pilot study served two purposes: (1) to assess the practicality and usability of the experimental procedure, and (2) to collect feedback for refining the study design.

One key issue identified during the pilot concerned how task details were presented. During each task, participants had to repeatedly click the corresponding button in the "Task List" sidebar to open a pop-up window containing the task details, such as the target address, token type, and amount. This interaction introduced unnecessary effort and disrupted the task flow.

To address this issue, we redesigned the interface to display task details in a fixed area at the top of the Task List page, so that the information remained visible throughout the task without requiring additional clicks. This change simplified the interaction flow and made task information more readily accessible, as illustrated in Appendix D.4.

D.3 Sample Size Determination

We conducted an *a priori* power analysis in G*Power 3.1 solely to determine the overall recruitment target for the experiment. For overall sample-size planning across all conditions, with transfer correctness as the primary outcome, we selected the ANOVA option in G*Power following practices reported in prior work [44]. [4] The analysis assumed a medium effect size (Cohen's $f = 0.25$), a significance level of $\alpha = 0.05$, and a desired statistical power of 0.95. Based on these parameters, G*Power estimated a required total

[4]For the actual analyses, we selected statistical methods appropriate to the measurement structure of each outcome.

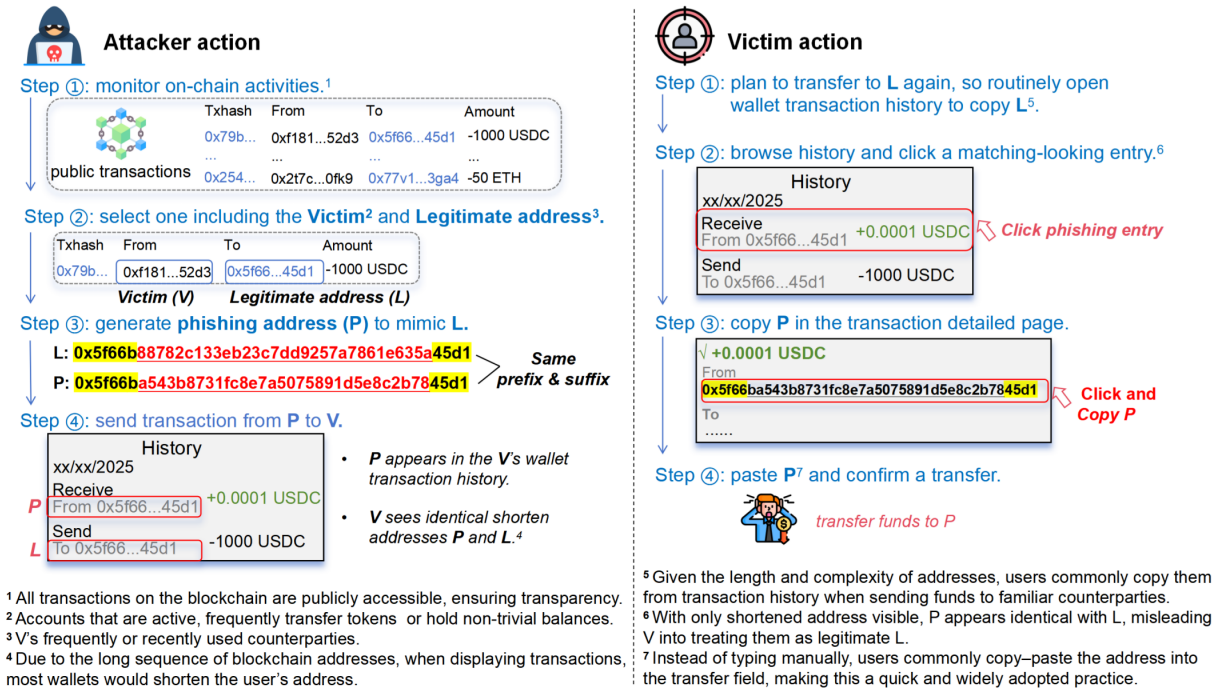


Figure 5: Basic mechanism of an address poisoning attack. Attacker steps ①-④ (left): Monitors on-chain activities, selects a victim, forges a visually similar address sharing the same prefix and suffix, and poisons the victim's transaction history. Victim steps ①-④ (right): Intends to pay a previously interacted recipient, browses history and clicks the matching-looking entry, copies and pastes the attack address, and mistakenly transfers funds to the attack address.

Table 2: Five experimental conditions.

Condition	Stage	Intervention
Control	–	Standard wallet interface; no intervention.
Visual Suppression	Transaction history	Fraudulent transaction shown as grayed-out rows with a "suspicious address" label.
Copy Warning	Address copying	Warning modal shown when copying a fraudulent address, prompting full-address verification.
Confirmation Disruption	Transaction submission	Warning modal shown when confirming a transfer to a fraudulent address, requiring explicit re-confirmation.
Combined	All stages	Combination of the three interventions above.

sample size of 305 participants, corresponding to approximately 61 participants per condition.

D.4 Online Experimental Platform Screenshot

The online experimental platform consisted of an instruction section (task instructions and task list) and a simulated wallet application (transaction selection, address copying, and transfer submission), as illustrated in Fig. 6.

D.5 Post-Test Survey

Details of the post-test survey, including condition-specific adaptations, are summarized below. Participants in the **Control** condition did not receive Q7–Q9. For participants in the **Visual Suppression** condition, Q7–Q9 were adapted as follows: Q7, "Did you notice the grayed-out records in the transaction history?"; Q8, "How did you understand why the record was grayed out?"; and Q9, "How helpful was the graying-out measure in helping you choose the correct address?" (1 = Not helpful at all, 5 = Very helpful).

- **Q1.** Please input your Prolific ID. *(Open-ended)*
- **Q2.** How confident are you about sending all transactions to the correct address? *(Likert scale, 1–5)*
- **Q3.** Which tasks did you find difficult? *(Open-ended)*
- **Q4.** Which information or elements helped you decide which address to choose? *(Open-ended)*
- **Q5.** Did you notice any suspicious transactions in the transaction history during the test? *(Binary choice: Yes/No)*
- **Q6.** What suspicious aspects did you notice in the suspicious transactions? *(Multiple choice)*
- **Q7.** Did you encounter the Risk Warning Popup? *(Binary choice: Yes/No)*
- **Q8.** Did you understand the content of the Risk Warning Popup? What actions did you take after seeing it? *(Open-ended)*
- **Q9.** How helpful was the Risk Warning Popup in helping you choose the correct address? *(Likert scale, 1–5)*

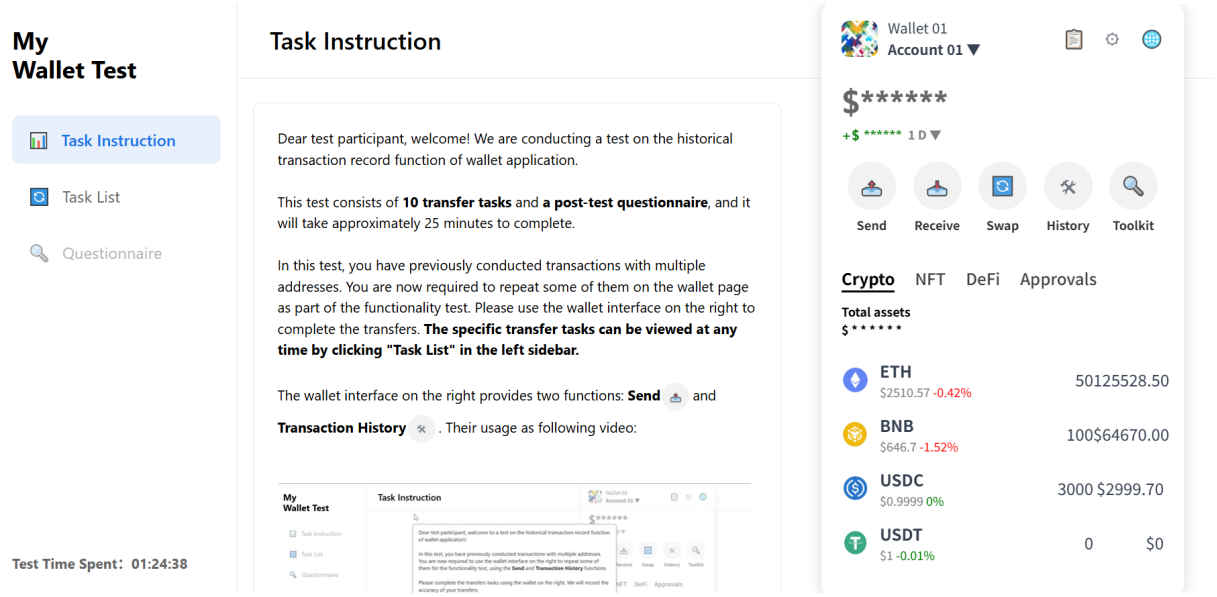


Figure 6: The online experimental platform, consisting of two main components: (1) Instruction section (the top two buttons on the left sidebar and central content panel). Clicking "Task Instruction", the central content panel presents study goals, an overview of the tasks, a wallet interface guide, and a short demonstration video; Clicking "Task List" causes the central panel to display details of the current task completion; (2) Simulated Wallet Application (right panel) enabling participants to select transaction, copy counterpart addresses, and send transfers.

- **Q10.** Did you compare the full 42-character address during this test? *(Multiple choice: frequency)*
- **Q11.** In what situations did you compare the full 42-character address? In what situations did you choose not to compare it? *(Open-ended)*
- **Q12.** Are you familiar with Web3? *(Binary choice: Yes/No)*
- **Q13.** How would you describe your understanding of Web3? *(Open-ended)*
- **Q14.** How long have you been using crypto wallet applications? *(Multiple choice: usage duration)*
- **Q15.** Which crypto wallets have you used? *(Open-ended)*
- **Q16.** Overall feedback: Do you have any suggestions or comments about the History and Send functions? *(Open-ended)*

E Supplementary Interview Materials

E.1 Interviewee Demographics

The demographics and condition assignments of the 20 participants in our semi-structured interviews are shown in Table 3.

Table 3: Participant demographics and condition assignments in the semi-structured interviews. *Note: Web3 Exp. indicates self-reported experience with crypto wallets. We refer to Visual Suppression, Copy Warning, Confirmation Disruption, and Combined Intervention as VS, CW, CD, and CI, respectively.*

ID	Occ.	G	Exp.	Cond.	ID	Occ.	G	Exp.	Cond.
P01	Student	M	1–3	CW	P11	Finance	F	1–3	CI
P02	Student	M	<1	CW	P12	Nurse	F	<1	CI
P03	Staff	M	1–3	CD	P13	Student	F	1–3	Control
P04	Operator	F	<1	CW	P14	Student	M	1–3	Control
P05	Manager	M	>3	VS	P15	Staff	M	1–3	CI
P06	Designer	M	<1	Control	P16	Engineer	F	<1	CW
P07	Operator	M	1–3	CD	P17	Student	M	1–3	CD
P08	Engineer	F	Never	VS	P18	Engineer	F	>3	CD
P09	Teacher	M	>3	CI	P19	Student	M	<1	VS
P10	Worker	M	1–3	VS	P20	Designer	F	<1	Control

E.2 Interview Protocol

Thank you very much for agreeing to participate in this interview. We are conducting a study to better understand how users respond to interventions in crypto wallets that are designed to mitigate address poisoning attack. In the following session, we will ask you a series of questions to learn more about how you made decisions during the experiment and how you experienced the interventions. This interview will be audio-recorded in full; however, your identity will remain strictly confidential, and your real name will never be used in any transcripts, reports, or public presentations. You may ask questions or pause the interview at any time. Your participation is entirely voluntary, and you may withdraw at any point without any negative consequences.

Do you consent to the recording of this interview?

Part I. Introduction and Background

- Are you familiar with address poisoning attacks?
- If yes:
  - Through what channels or sources did you learn about them?
  - How do you understand address poisoning attacks?
- If no:

- Provide a brief explanation of the mechanism of address poisoning attacks.

Part II. Decision-Making Process

- How did you decide which transaction to click on for the current task when selecting from the transaction history?
- When selecting addresses, did you compare the full 42-character address?
  - If yes:
    - Why did you choose to compare the full address?
  - If no:
    - Why did you not compare the full address?
    - What obstacles or difficulties prevented you from doing so?
  - Did you instead compare only part of the address? Why or why not?
- When selecting from the transaction history, did you pay attention to any information other than the address?
  - If yes:
    - What information did you notice?
    - For what purpose did you pay attention to this information?
    - Did it help you decide which address to select?
  - If no:
    - Why did you not pay attention to other information?
- What information did you check when copying addresses from the detail page?
  - For what purpose did you verify this information?
  - Did you compare the full 42-character address? Why or why not?
- What information did you verify when confirming transactions?
  - For what purpose did you verify this information?
  - Did you compare the full 42-character address? Why or why not?

For participants in the Visual Suppression condition:

- Did you notice the grayed-out entries?
  - If yes:
    - How did you interpret them?
    - Did they affect your choice? Why?
    - How did you understand the label "suspicious address"?

For participants in the Copy Warning condition:

- Did you notice the popup warning when copying an address on the detail page?
  - If yes:
    - How did you interpret it?
    - Did it affect your choice? Why?
  - Did you follow the instruction to recheck the full 42-character address? Why or why not?
    - How did you understand the term "suspicious address"?

For participants in the Confirmation Disruption condition:

- Did you notice the popup warning when confirming a transaction on the detail page?
  - If yes:
    - How did you interpret it?
    - Did it affect your choice? Why?
  - Did you follow the instruction to recheck the full 42-character address? Why or why not?
    - How did you understand the term "suspicious address"?

Part III. Comparative Perspectives

- In traditional financial systems, do you check the full account identifier before sending money?
- What differences do you perceive between traditional transfers and Web3 transfers?
- How do these differences influence your caution or verification strategies?

Part IV. Feedback and Suggestions

- How helpful did you find the graying-out (for participants in the **Visual Suppression** condition) or popup warnings (for participants in the **Copy Warning** or **Confirmation Disruption** condition)?
- What limitations or shortcomings did you notice in these interventions?
- Do you have any other comments or suggestions?

E.3 Codebook

Figure 7 displays the final codebook for interview analysis.

F Supplementary Findings

F.1 Overview of Findings

Figure 8 provides an overview summary of the effectiveness of the three nudges, including correctness improvements, statistical results, behavioral insights, and heterogeneity analyses.

F.2 Illustrative Example: Impact of Confirmation Disruption on Participant Task Flow

Figure 9 illustrates an example user flow, showing how the warning pop-up (*in the Confirmation Disruption*) shaped a participant's behavior during transfer.

F.3 Heterogeneity Analysis

Beyond evaluating overall effectiveness, we also examine how contextual and individual differences shape users' responses to address poisoning attacks. In particular, we analyze whether prior Web3 experience, repeated exposure to attack attempts, and task-specific factors influence users' susceptibility to attack and their behavioral responses to the interventions.

F.3.1 Individual Differences by Web3 Experience. We examined whether prior crypto wallet experience influenced users' susceptibility to address poisoning attacks. Participants were grouped into four self-reported experience levels: none, less than one year, 1–3 years, and more than 3 years, and their experimental condition was recorded. Task correctness was modeled as a function of condition assignment and experience level using GEE logistic regression, treating experience as a categorical predictor to test for its main effect (across all conditions) and its interaction with experimental conditions. We also analyzed behavioral outcomes such as `FirstClickSafe`, `CopyChange`, and `ConfirmModalChoice`. As shown in Table 11, we found no significant effects of Web3 experience. The overall main effect of experience across all conditions was not significant, correctness in the *Control* condition did not vary significantly across the four levels, and no interaction between

Part1.Background	Avoid click	Usability barriers	Usability barriers
A. Understanding	B.2.3 Design perception	D.When confirming transactions	E.Partial address matching
Never heard	Less trustworthy entries	D.1 Information focus	Complex string
Heard	C.When copying addresses	Amount Correctness	Wallet trust
Personally experienced	C.1 Information focus	Token Type Correctness	Limited phishing awareness
Part2.Decision-Making Process	Source position (sender/receiver)	First-last few-characters check	**Part III. Traditional financial compari**
B. During browsing tansaction	First-last few-characters check	D.2 Comfirmation Distruption design	F.Full comparison
B.1 Information focus	C.2 Copy Warning design	D.2.1 Design interpretation	Simple string
Normal small amounts	C.2.1 Design interpretation	Suspicious signal	Readable characters
First-last few-characters check	Suspicious signal	Routine prompt	G.Partial comparison
B.2 Visual Suppresion design	Routine prompt	D.2.2 Action affect	Institutional safeguards
B.2.1 Design interpretation	C.2.2 Action affect	D.2.2.1 Ignore	**Part IV. Feedback and Suggestions**
Suspicious signal	C.2.2.1 Ignore	No urgency warning	H.Clearer Guidance
Canceled transaction	Low-risk perception	D.2.2.2 Recheck	Clearer reasons
Failed transaction	C.2.2.2 Recheck	Risk perception	Specific checking parts
Outdated transaction	Risk perception	Final/irreversible stage	
B.2.2 Action affect	C.2.2.3 Recheck but failed	D.2.2.3 Recheck but failed	
Click but drop	Superficial verification strategies	Superficial verification strategies	

Figure 7: The final codebook.

experience and intervention reached significance ($p > 0.05$). Similarly, models of stage-specific behaviors showed no main effects of experience level ($p > 0.05$).

To interpret this lack of variation, we analyzed interview responses and summarized two reasons. First, participants, regardless of experience, relied on similar heuristics when judging address legitimacy, typically comparing only the beginning and end of the string (Section 5.2.2 and Section 5.2.3). Second, participants exhibited generally low awareness of address poisoning as an attack vector, tending to trust any address shown in the wallet interface. As P06 explained: *"When transferring to someone I know, I think it's definitely correct because I'd used the address several times before. It never occurred to me that the middle part of the address could be different"* (P06).

F.3.2 Behavioral Adaptation over Repeated Exposure. Our experiment introduced repeated attack attempts across multiple tasks. This allowed us to examine whether participants adapted over time—either by becoming more responsive to nudges (learning) or by becoming less responsive over repeated exposure.

We focused on participants in the *CW*, *CD* and *CI* conditions who triggered warning modals. Each participant's responses across exposures were tracked, and both immediate modal choices (`ModalChoice`) and outcomes (`Correctness`) were modeled using GEE logistic regression with participant ID clustered. Results showed a consistent decline in responsiveness over repeated exposures (Table 12): in the *CW* ($p < 0.05$), *CD* ($p < 0.05$) and *CI* conditions ($p < 0.05$), the probability of returning to recheck declined significantly with repeated exposures. Task-learning effects appear less likely to explain this decline, as `TriggerIndex` was not significantly associated with correctness ($p > 0.05$). The post-test survey provided insight: in CI, some participants considered earlier verification steps sufficient and perceived the subsequent confirmation warning as unnecessary. Others reported less checking after repeated exposure when previous checks revealed no issues. These observations suggest that reduced responsiveness in *CI* may reflect both warning redundancy from earlier interventions and habituation. Meanwhile, the declines in *CW* and *CD* are more consistent with habituation, as participants became less likely to recheck after repeated warnings.

Importantly, once participants returned to recheck, the quality of their verification remained stable. That is, users who paused were just as likely to submit the correct address later in the study as they were earlier ($p > 0.05$ in both conditions). This indicates that repeated exposure reduced the frequency of compliance but not the effectiveness of compliance once initiated.

F.3.3 Content Differences of Malicious Transactions. Finally, we examined whether variations in attack content, specifically transaction type, token amount and token type, affected user behavior. To mitigate order effects, the sequence of ten transfer tasks was randomized across participants so that each participant encountered a unique task order.

Task correctness was modeled as a function of attack type using GEE logistic regression clustered by participant, with separate analyses by intervention condition. Results (Table 13) showed little systematic variation: across all conditions, no specific attack variant (e.g., different-token transfers, or multiple injections) significantly predicted correctness ($p > 0.05$). In interviews, only a few participants reported noticing transaction details beyond the address itself, and even when differences were observed, they were dismissed as unremarkable. As P13 noted: *"I thought it was normal, my friends also often send me small test transfers"* (P13).

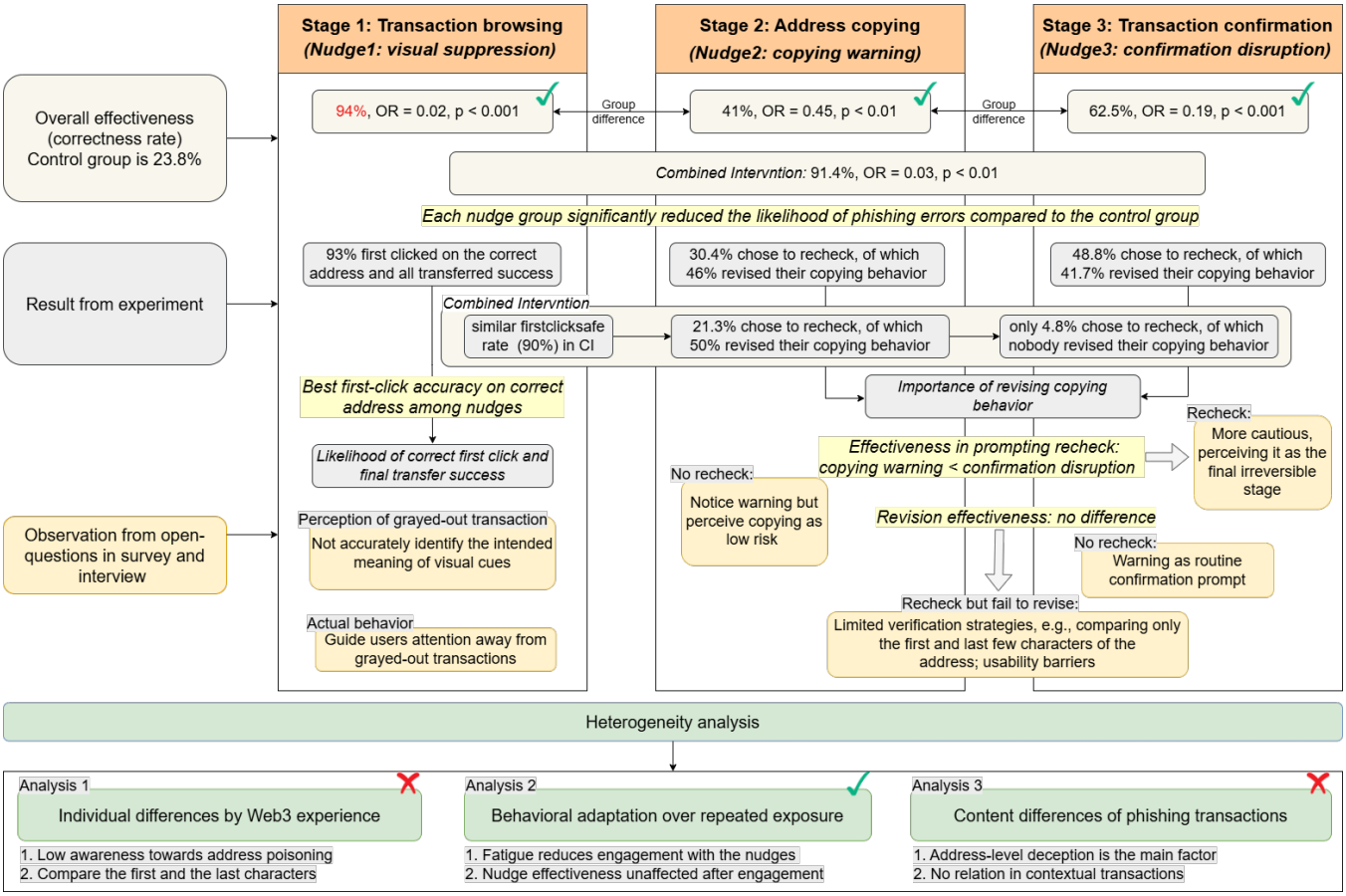


Figure 8: Summary of findings regarding the effectiveness of three nudges. The first row (light yellow) compares task correctness between each nudge and the Control condition, showing that all nudges improved performance. The second row (light gray) reports statistical results and key findings, while the third row (light orange) highlights behavioral insights under each intervention. The fourth row (green) summarizes heterogeneity analyses. Yellow-highlighted text marks comparative conclusions across different interventions.

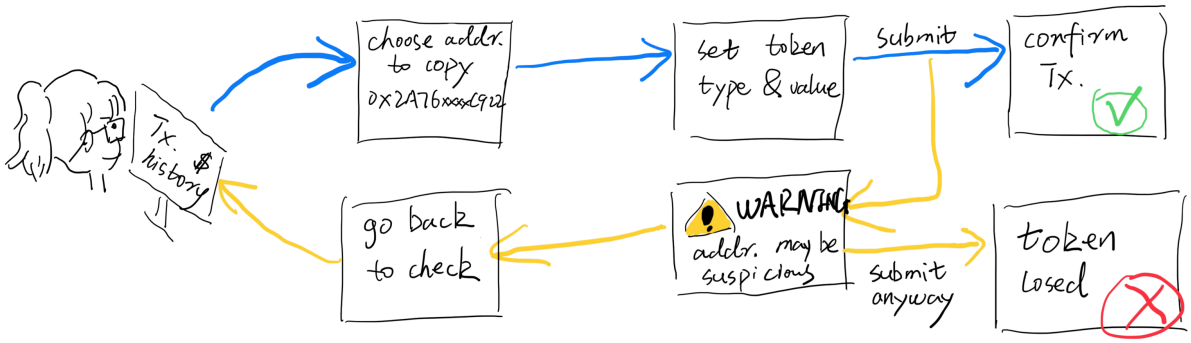


Figure 9: Illustration of how encountering a warning pop-up (in the Confirmation Disruption condition) shaped a participant (P18)'s task flow, highlighting how such interventions helped users complete transactions in the context of an address poisoning attack.

Table 4: Pairwise comparisons between experimental conditions. The dependent variable is `correct` (1 = transaction sent to the legitimate address, 0 = to the fraudulent address). Odds ratios (OR) therefore represent the relative odds of success (`correct=1`) between conditions: OR > 1 indicates that the first condition was more likely to complete transactions correctly than the second, whereas OR < 1 indicates a lower likelihood of success. Risk differences (RD) reflect the absolute change in success rates. Wald χ^2 values and p-values indicate the statistical significance of the differences.

Comparison	OR [95% CI]	RD [95% CI]	Wald χ^2	p-value
Control vs Visual Suppression	0.02 [0.01, 0.04]	-0.700 [-0.742, -0.659]	87.97	6.64×10^{-21}
Control vs Copy Warning	0.45 [0.28, 0.73]	-0.170 [-0.227, -0.114]	10.74	0.0011
Control vs Confirmation Disruption	0.19 [0.11, 0.31]	-0.384 [-0.441, -0.327]	40.44	2.03×10^{-10}
Control vs Combined Intervention	0.03 [0.01, 0.06]	-0.672 [-0.716, -0.629]	98.58	3.12×10^{-23}
Visual Suppression vs Copy Warning	22.86 [9.85, 53.07]	0.530 [0.481, 0.578]	53.01	3.33×10^{-13}
Visual Suppression vs Confirmation Disruption	9.51 [4.01, 22.52]	0.316 [0.267, 0.366]	26.14	3.18×10^{-7}
Visual Suppression vs Combined Intervention	1.53 [0.58, 4.06]	0.029 [-0.004, 0.062]	0.72	0.395
Copy Warning vs Confirmation Disruption	0.42 [0.24, 0.72]	-0.214 [-0.276, -0.151]	9.62	0.0019
Copy Warning vs Combined Intervention	0.07 [0.03, 0.14]	-0.502 [-0.552, -0.451]	54.18	1.83×10^{-13}
Confirmation Disruption vs Combined Intervention	0.16 [0.08, 0.34]	-0.287 [-0.339, -0.236]	23.17	1.48×10^{-6}

Table 5: Results for the question in the survey: *"How confident are you about sending all transactions to the correct address?"* Descriptive statistics (Mean, SD, Median) across five conditions. A Kruskal–Wallis test indicated no significant differences in confidence across conditions.

Condition	Mean (M)	SD	Median
Control	4.62	0.63	5.0
Visual Suppression	4.69	0.70	5.0
Copy Warning	4.61	0.60	5.0
Confirmation Disruption	4.50	0.76	5.0
Combined Intervention	4.73	0.51	5.0
Kruskal–Wallis test: $p = 0.2919$			

Table 6: Pairwise comparisons of `FirstClickSafe` between the Visual Suppression condition and other conditions. p-values are obtained from Wald χ^2 tests.

Comparison (vs. Visual Suppression)	p
Control	1.98×10^{-26}
Copy Warning	3.01×10^{-23}
Confirmation Disruption	1.54×10^{-20}
Combined Intervention	0.403

F.4 Statistical Results

Tables 4–13 provide statistical results, including detailed pairwise comparisons between experimental conditions, transaction confidence, behavioral mechanisms, and heterogeneity analyses on crypto wallet experience, changes in responsiveness over repeated exposures, and fraudulent content effects. In all tables in this section, $p < 0.05$ indicates a statistically significant difference.

Table 7: Comparison of `CopyChange` rates between the Copy Warning, Control and Combined Intervention conditions. CW stands for Copy Warning, and CI for Combined Intervention. Wald χ^2 tests were used to assess condition differences.

Condition	CopyChange = 1	CopyChange = 0
Control	7%	93%
CW	14%	86%
CI	10.6%	89.4%
Control vs. CW: $p = 0.0183$		
CI vs. CW: $p = 0.9812$		

Table 8: Distribution of modal responses (`ModalChoice`) across conditions, presented as percentages with Wilson 95% confidence intervals. CW stands for Copy Warning, CD for Confirmation Disruption, and CI for Combined Intervention. p-values correspond to Wald χ^2 tests comparing the proportion of "Return" responses between conditions.

Condition	Proceed	Return
CW	69.6 [64.0, 76.3]	30.4 [23.7, 36.0]
CD	51.2 [44.5, 57.8]	48.8 [42.2, 55.5]
CI	95.2 [84.2, 98.7]	4.8 [1.3, 15.8]
CW vs. CD: $p = 0.036$		
CI vs. CD: $p = 7.36 \times 10^{-4}$		

Table 9: Modal response patterns. (N) is the number of participants who reached the CD modal at least once. Percentages are sequentially conditioned on modal reach and, subsequently, an initial "ReCheck" response.

Condition	First ReCheck	Continue ReCheck
CI ($N = 13$)	15.4%	0.0%
CD ($N = 56$)	76.8%	83.7%

Table 10: Correctness rates among participants who chose *Return*. CW stands for Copy Warning, CD for Confirmation Disruption, and CI for Combined Intervention. The p-value corresponds to a Wald χ^2 test comparing such rates.

Condition	Correct	Incorrect
CW.R	46 [32.5, 56.7]	54 [43.3, 67.5]
CD.R	41.7 [32.7, 51.4]	58.3 [48.6, 67.3]
CI.R	0.0 [0.0, 65.8]	100.0 [34.2, 100.0]
CW.R vs. CD.R: $p = 0.880$		

Table 11: Experience main effects (Control baseline), Group × Experience interaction, and Stage-specific behaviors. All reported values are p-values from Wald χ^2 tests. Experience levels correspond to four self-reported categories: none (Exp1), less than one year (Exp2), 1–3 years (Exp3), and more than 3 years (Exp4).

Group	Exp2	Exp3	Exp4
Visual Suppression	0.776	0.056	0.591
Copy Warning	0.951	0.086	0.671
Confirmation Disruption	0.309	0.320	0.586
Combined Intervention	0.431	0.952	0.797
Behavior	**Exp2**	**Exp3**	**Exp4**
FirstClickSafe	0.361	0.243	0.752
CopyChange	0.097	0.063	0.501
ConfirmModalChoice	0.710	0.338	0.722
Experience main effects			
Exp2		0.282	
Exp3		0.443	
Exp4		0.385	

Table 12: Changes in participants' responses across repeated exposures to warning modals. Combined Intervention 1 stands for response to Copy Warning in Combined Intervention, while Combined Intervention 2 stands for that to Confirmation Disruption. Negative β values indicate declining likelihood of returning over time.

Condition	β	p
Copy Warning	-0.513	1.7×10^{-5}
Confirmation Disruption	-0.335	4×10^{-6}
Combined Intervention 1	-0.594	0.0337
Combined Intervention 2	-1.684	0.0401

Table 13: Wald χ^2 tests for the effect of fraudulent content on correctness. p-values represent the joint significance of all fraudulent content within the condition.

Condition	p
Control	0.567
Visual Suppression	0.489
Copy Warning	0.192
Confirmation Disruption	0.073
Combined Intervention	0.383